

2023-2024
 2nd year /Electronic I
 Electronic and Telecommunication Department.
 Engineering Science/ Kufa University

Single Stage Integrated Circuit MOSFET Amplifiers

4.1: The loading lines of the single transistor with resistive load

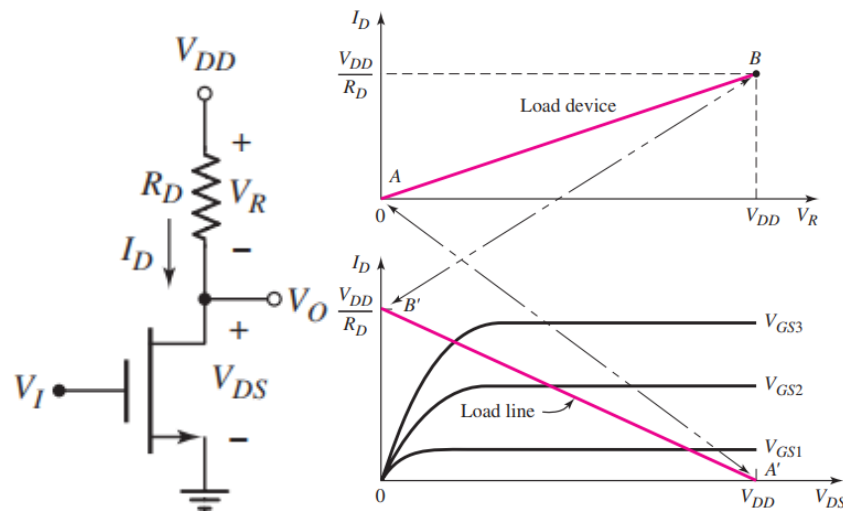


Figure 4.1 Single MOSFET circuit with resistive load and the I–V curve

4.2: NMOS Amplifiers with Enhancement Load

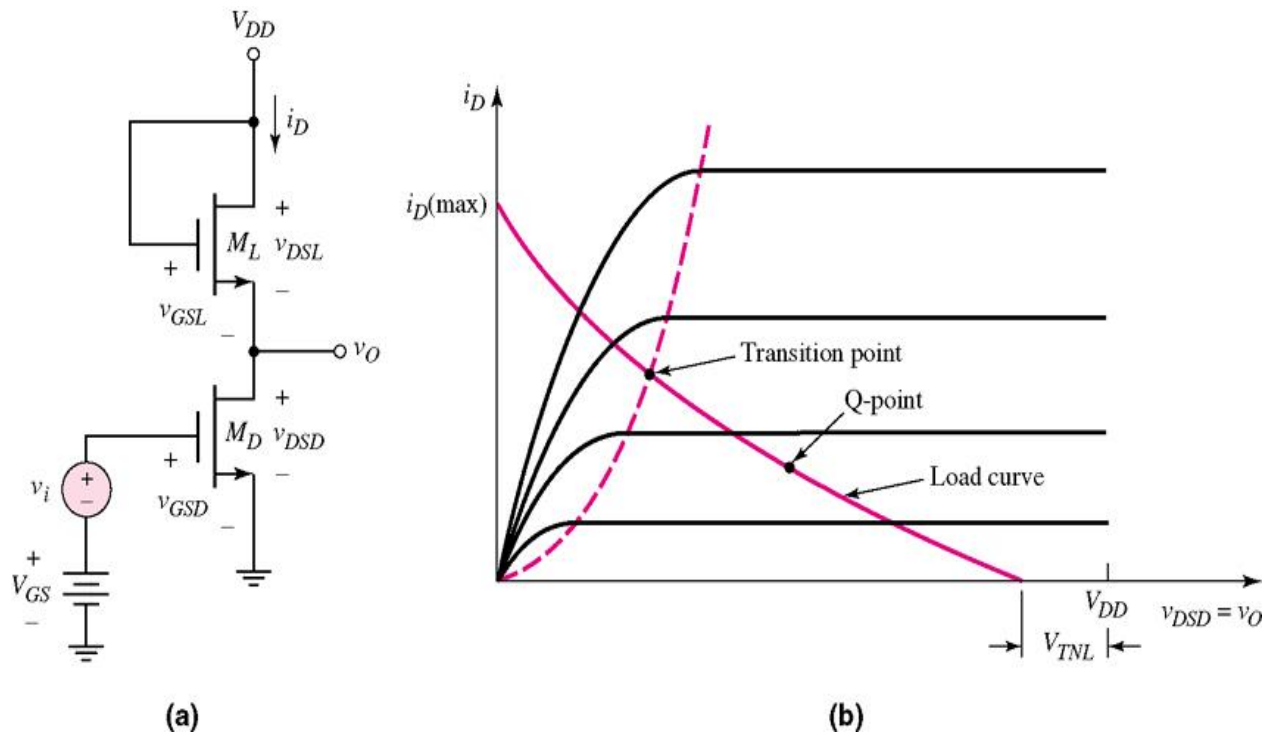


Figure 4.2 NMOS amplifier with enhancement load device

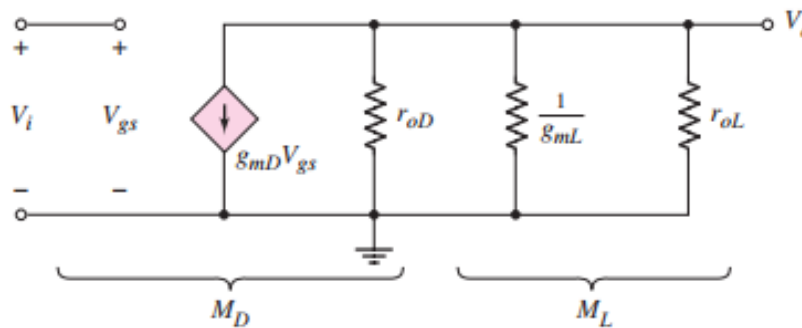


Figure 4.40 Small-signal equivalent circuit of NMOS inverter with enhancement load device

The small-signal voltage gain is then

$$A_v = \frac{V_o}{V_i} = -g_{mD} \left(r_{oD} \parallel \frac{1}{g_{mL}} \parallel r_{oL} \right)$$

generally, $1/g_{mL} \ll r_{oL}$ and $1/g_{mD} \ll r_{oD}$, the voltage gain

$$\approx -\frac{g_{mD}}{g_{mL}} = -\sqrt{\frac{K_{nD}}{K_{nL}}}$$

Then at $k'_{nD} = k'_{nL}$

$$= -\sqrt{\frac{(W/L)_D}{(W/L)_L}}$$

The voltage gain, then, is related to the size of the two transistors.

$$\approx -\frac{g_{mD}}{g_{mL}} = -\sqrt{\frac{K_{nD}}{K_{nL}}} = -\sqrt{\frac{(W/L)_D}{(W/L)_L}}$$

4.3: NMOS amplifier with depletion load

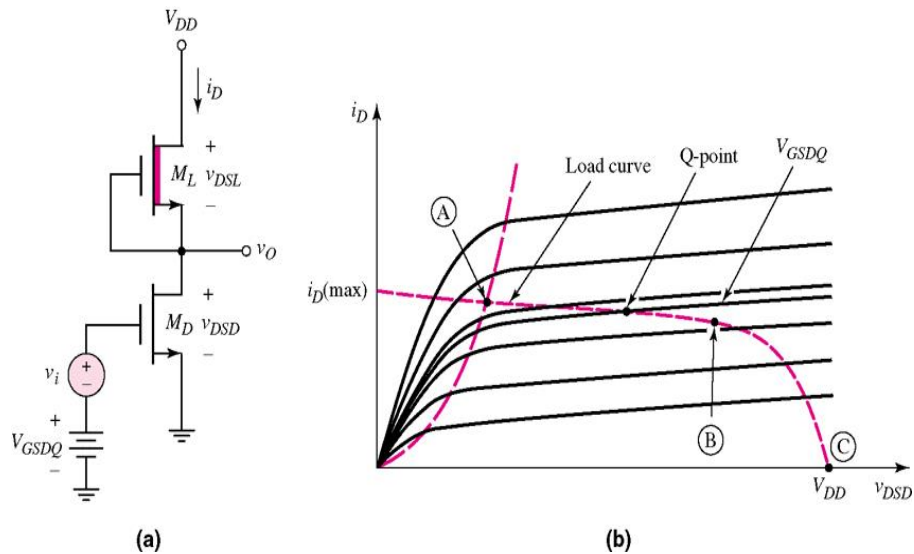


Figure 4.3 (a) NMOS amplifier with depletion load device with driver transistor characteristics

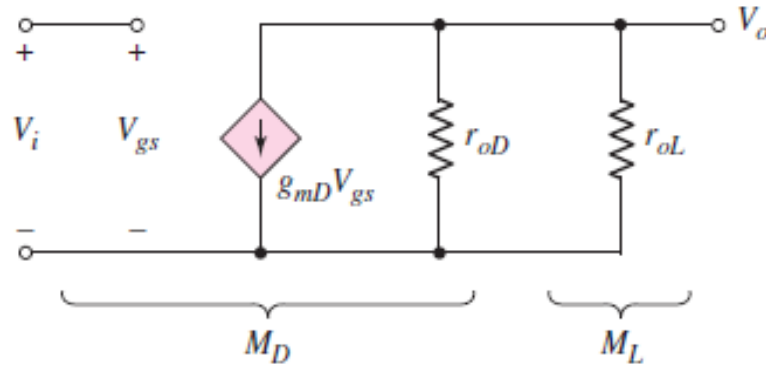


Figure 4.4 Small-signal equivalent circuit of NMOS inverter with depletion load device

$$A_v = \frac{V_o}{V_i} = -g_{mD}(r_{oD} \parallel r_{oL})$$

4.4. NMOS amplifier with active loads

4.4.1: CMOS common source amplifier

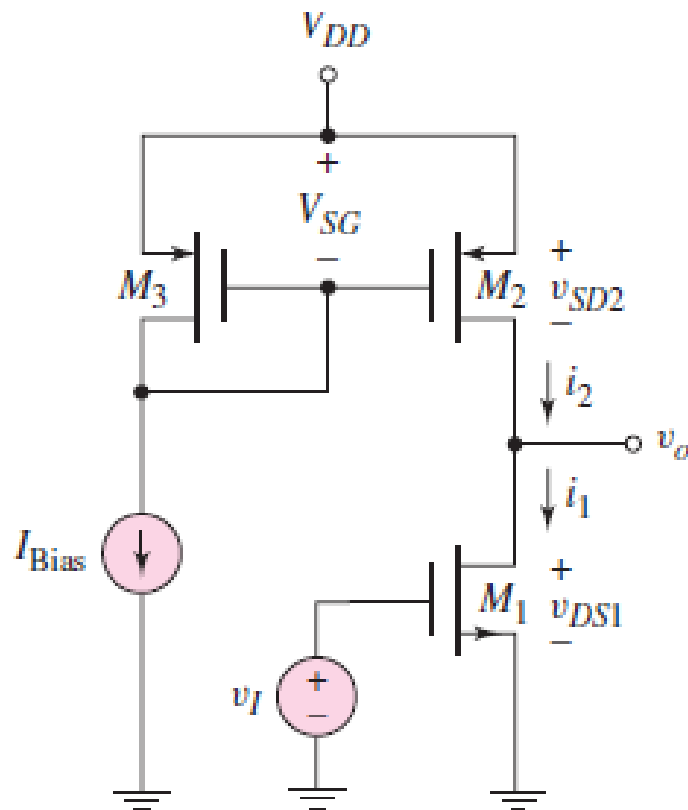


Figure 4.5 CMOS common-source amplifier

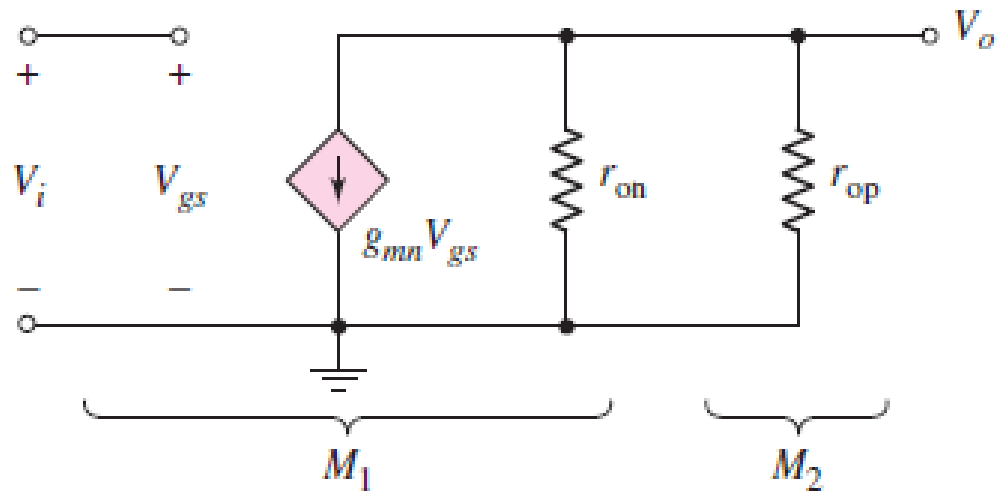


Figure 4.6 Small-signal equivalent circuit of the CMOS common-source amplifier

$$A_v = \frac{V_o}{V_i} = -g_{mn}(r_{on} \parallel r_{op})$$

4.4.2: CMOS Source-Follower Amplifier

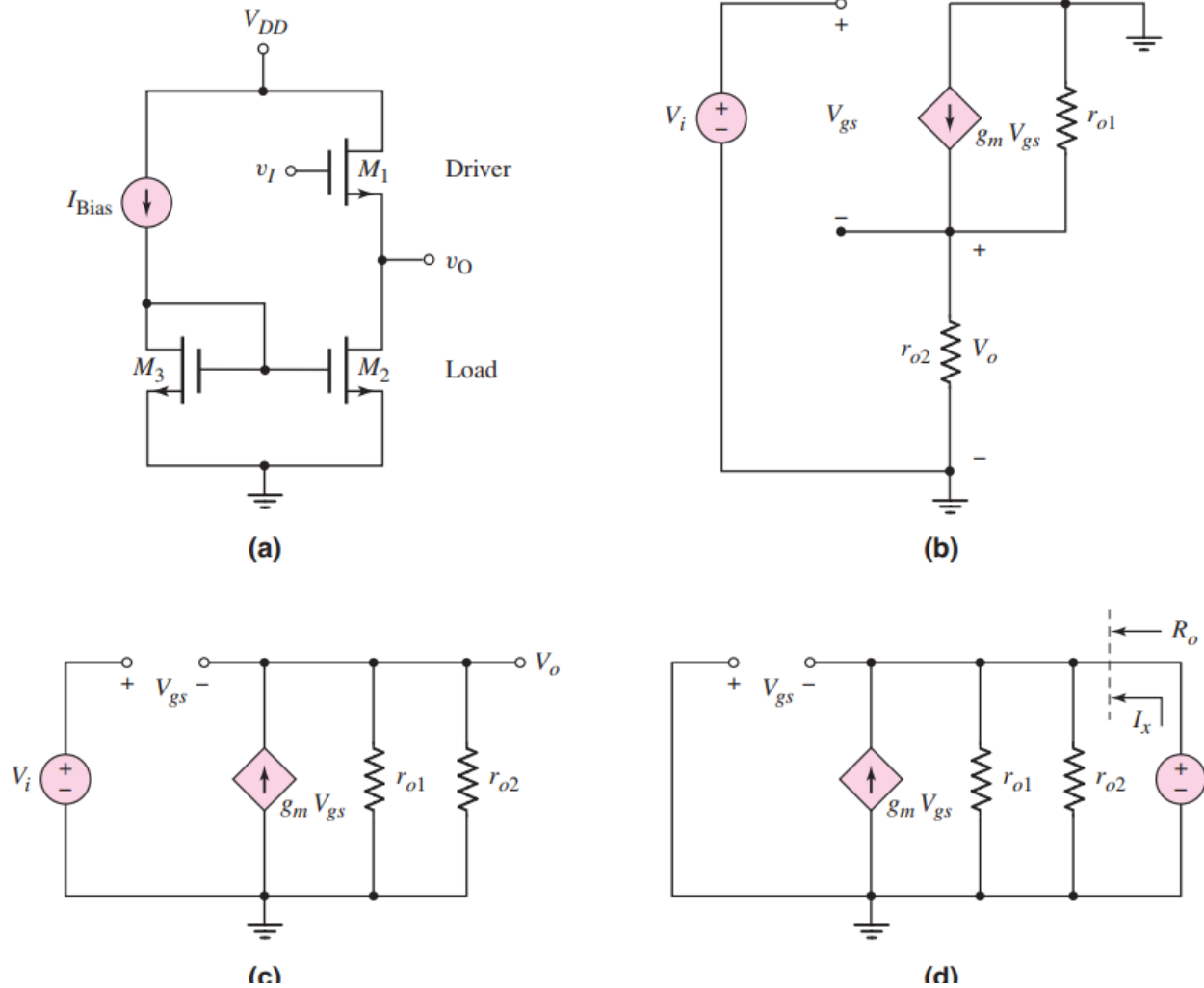


Figure 4. 7: (a) All NMOS source-follower circuit, (b) small-signal equivalent circuit, (c) reconfiguration of small-signal equivalent circuit, and (d) small-signal equivalent circuit determine the output resistance

$$A_v = \frac{V_o}{V_i} = \frac{g_{m1}(r_{o1} \parallel r_{o2})}{1 + g_{m1}(r_{o1} \parallel r_{o2})}$$

$$R_o = \frac{V_x}{I_x} = \frac{1}{g_{m1}} \parallel r_{o2} \parallel r_{o1}$$

4.4.3: CMOS common gate amplifier

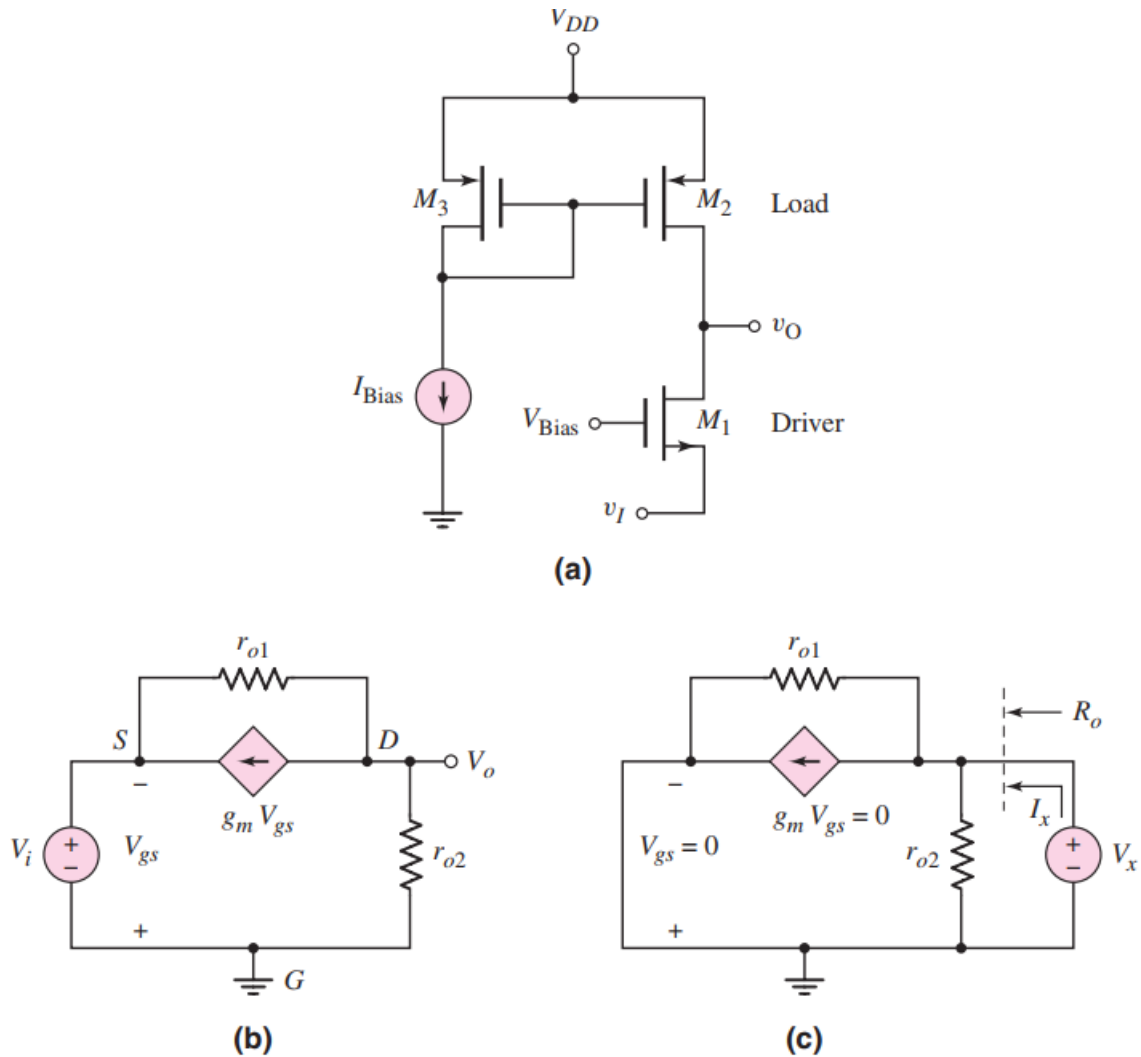


Figure 4.8 (a) CMOS common-gate amplifier, (b) small-signal equivalent circuit, and (c) small-signal equivalent circuit for determining the output resistance

$$\frac{V_o}{r_{o2}} + g_{m1} V_{gs} + \frac{V_o - (-V_{gs})}{r_{o1}} = 0$$

$$V_o \left(\frac{1}{r_{o2}} + \frac{1}{r_{o1}} \right) + V_{gs} \left(g_{m1} + \frac{1}{r_{o1}} \right) = 0$$

$$A_v = \frac{\left(g_{m1} + \frac{1}{r_{o1}} \right)}{\left(\frac{1}{r_{o2}} + \frac{1}{r_{o1}} \right)}$$

$$I_x = \frac{V_x}{r_{o2}} + g_{m1} V_{gs} + \frac{V_x - (-V_{gs})}{r_{o1}}$$

However, $V_{gs} = 0$ so that $g_{m1} V_{gs} = 0$. We then find

$$R_o = \frac{V_x}{I_x} = r_{o1} \parallel r_{o2} = 500 \parallel 500$$

4.5: Multistage Amplifier

4.5.1: Cascade Circuit:

In most applications, a single-transistor amplifier will not be able to meet the combined specifications for a given amplification factor, input impedance, and output impedance.

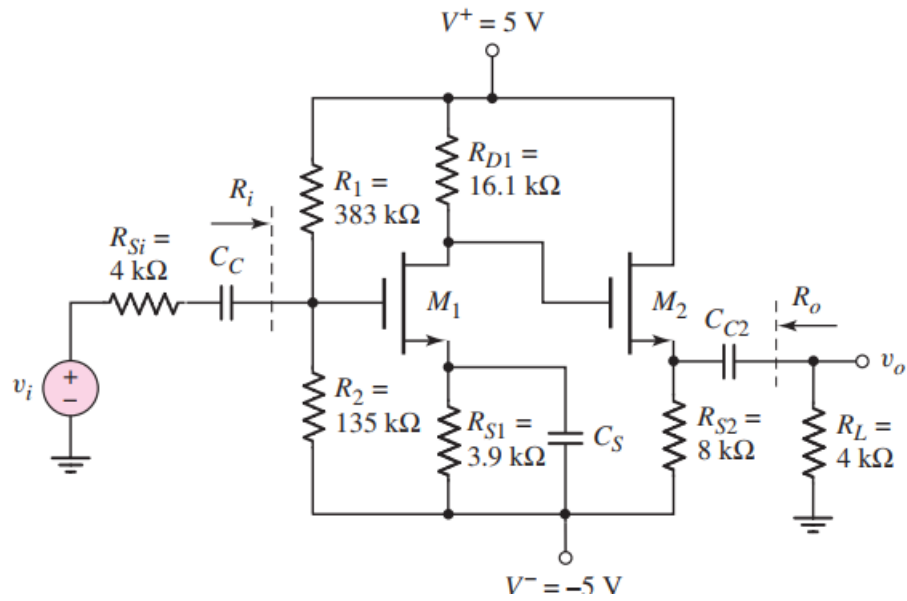


Figure 4.9 Common-source amplifier in cascade with source follower

4.5.2: Cascode Circuit

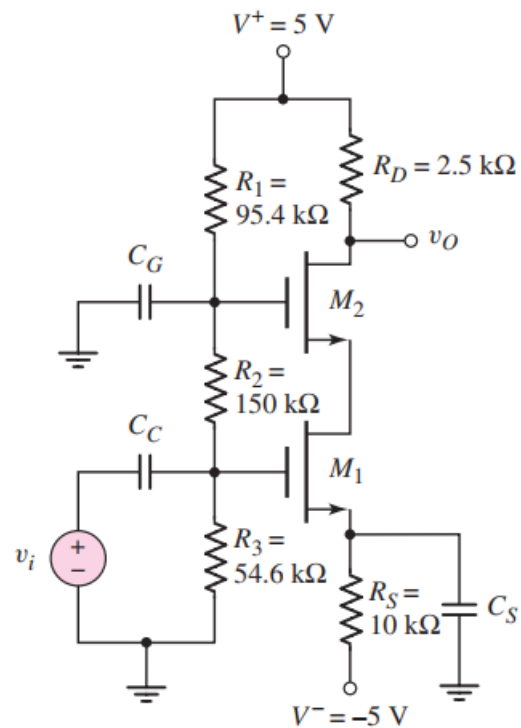


Figure 4.10 NMOS cascode circuit